

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 5

To be covered during the examples class at 09:00 on Thursday 30th April. These exercises do not count toward your final module mark, however it is a good idea to attempt these questions prior to the examples class.

1. (a) For the general ARMA process, $\phi(B)X_t = \theta(B)Z_t$, deduce that the spectral density function is given by

$$g(w) = \frac{\sigma_z^2}{2\pi} \frac{\theta(e^{-iw})\theta(e^{iw})}{\phi(e^{-iw})\phi(e^{iw})}, \quad -\pi < w \leq \pi.$$

- (b) Hence verify the form of $g(w)$ given in lectures for a stationary AR(2) process.

- (c) Show also that for a (weakly) stationary ARMA(1, 1) process

$$(1 - \phi B)X_t = (1 + \theta B)Z_t$$

we have

$$g(w) = \frac{\sigma_z^2}{2\pi} \frac{\{1 + \theta^2 + 2\theta \cos(w)\}}{\{1 + \phi^2 - 2\phi \cos(w)\}}, \quad -\pi < w \leq \pi.$$

2. A linear filter, applied to the stationary process $\{Y_t\}$, is defined

$$X_t = Y_t - Y_{t-4}$$

- (a) Determine the transfer function for this filter and sketch the squared gain.
 (b) Hence discuss the effect of the filter.

3. Suppose a linear filter is defined by

$$X_t = Y_t - 2Y_{t-1} \cos(\bar{\omega}) + Y_{t-2},$$

where $\bar{\omega}$ is a fixed frequency.

- (a) Determine the form of the transfer function for this filter.
 (b) For which frequencies ω does the spectral density function of $\{X_t\}$ equal zero?

4. It is known that the spectral density function of a certain MA(2) process $\{X_t\}$ is given by

$$g(\omega) = \frac{1}{2\pi} \{19 + 5 \cos(\omega) - 6 \cos(2\omega)\}.$$

If $X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$, where $\{Z_t\}$ is a white noise process with variance σ_z^2 , show that

$$1 + \theta_1^2 + \theta_2^2 = \frac{19}{\sigma_z^2}, \quad 2\theta_1(1 + \theta_2) = \frac{5}{\sigma_z^2} \quad \text{and} \quad \theta_2 = -\frac{3}{\sigma_z^2}.$$

5. The process $\{X_t\}$ is an $ARMA(1, 2)$ process

$$X_t = \phi_1 X_{t-1} + Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 and $|\phi| < 1$, $|\theta_1| < 1$ and $|\theta_2| < 1$. Determine the spectral density function of this process.

6. Suppose that $\{X_t\}$ and $\{Y_t\}$ are independent stationary processes with mean zero and autocovariance functions $\gamma_X(h)$ and $\gamma_Y(h)$, respectively. In addition, the spectral density functions of $\{X_t\}$ and $\{Y_t\}$ are $g_X(w)$ and $g_Y(w)$, respectively.

The process $\{V_t\}$ is such that

$$V_t = X_t + Y_t.$$

- (a) Show that the process $\{V_t\}$ is also stationary and has spectral density function $g_V(w)$ such that

$$g_V(w) = g_X(w) + g_Y(w).$$

- (b) Suppose that $\{X_t\}$ is an $AR(1)$ process, defined

$$X_t = \phi X_{t-1} + W_t$$

where $|\phi| < 1$ and $\{W_t\}$ is a white noise process with variance σ^2 . In addition, $\{Y_t\}$ is a white noise process with variance σ^2 . Determine the spectral density function of

$$V_t = X_t + Y_t.$$

7. (a) Derive the spectral density functions of

(i) An MA(1) process $X_t = Z_t + \theta Z_{t-1}$.

(ii) The process $\{X_t\}$ where

$$(1 - \rho B)^2 X_t = Z_t$$

with $|\rho| < 1$.

Here $\{Z_t\}$ is a white noise process with variance σ_z^2 .

(b) Suppose that $\{X_t\}$ is the linear filter

$$X_t = \frac{2}{3}Y_t + \frac{1}{3}Y_{t-1}$$

and $\{Y_t\}$ is a stationary process with mean μ and spectral density function $g_Y(w) = \sigma_z^2(1 - \beta \cos w)$. Derive the mean and spectral density function of $\{X_t\}$.

8. The process $\{X_t\}$ is the AR(1) process

$$X_t = \frac{1}{2}X_{t-1} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 . Suppose that $\{X_t\}$ is smoothed using the following linear filter (a moving average on three points) where

$$Y_t = \frac{1}{3}[X_{t+1} + X_t + X_{t-1}].$$

(a) Determine the spectral density function of $\{Y_t\}$.

(b) Sketch the spectral density function and squared gain for values of the frequency, w , where $w \in [0, \pi]$.